

Rupam Pal

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SUMMARY

AI Engineer and Full Stack Developer with **3 years of experience** building and deploying AI-powered applications and enterprise web platforms. Hands-on expertise in **RAG pipelines, Agentic AI** and **multi-agent orchestration** using **LangChain** and **LangGraph**, **LLM API integration (Groq, Gemini, OpenAI)**, **prompt engineering**, and **multimodal AI** — applied at production scale at Euromonitor International. Full-stack proficiency in **Angular, React, TypeScript, ASP.NET Core**, and **FastAPI**, backed by strong **Microsoft Azure** cloud, **AKS, Docker**, and **Azure Pipelines** CI/CD experience. Certified **AZ-204 Azure Developer Associate**.

TECHNICAL SKILLS

AI / ML: LangChain, LangGraph, LLM APIs (Groq, Gemini, OpenAI), RAG Pipelines, Multi-Agent Orchestration, Prompt Engineering, ML.NET

Languages: Python, C#, TypeScript, JavaScript, Java

Backend: FastAPI, ASP.NET Core, .NET Core 8.0, MVC Architecture, Web API, RESTful APIs, Entity Framework Core, LINQ, CQRS, Node.js

Frontend: React, Angular, React Router, HTML5, CSS3, jQuery, Tailwind CSS, TypeScript, Vite, Responsive Web Design

Cloud / DevOps: Microsoft Azure, Azure App Services, AKS, Azure DevOps, Azure Pipelines, Docker, Helm, CI/CD, Render, Vercel

Auth / Security: Microsoft Entra ID, Azure Active Directory, OAuth 2.0, OpenID Connect, RBAC, Azure Key Vault

Databases: SQL Server, PostgreSQL, MySQL, T-SQL, Stored Procedures, Query Optimization, MongoDB

Testing / Tools: Cypress, Postman, Swagger, Scalar, Git, GitHub, Agile, Scrum

WORK EXPERIENCE

EUROMONITOR INTERNATIONAL | SDE-2 | AI ENGINEER & FULL STACK DEVELOPER Hybrid | Apr 2025 – Present

- Engineered an **Agentic AI** sprint analytics platform integrating **ML.NET FastForest** classification with **Generative AI** LLM APIs (**Groq Llama 3.3, Google Gemini 2.5 Flash**), achieving 97.8% real-time risk prediction accuracy and delivering **AI-generated**, role-specific recommendations for Developers, BAs, PMs, and Scrum Masters.
- Designed and delivered the **North Star Delivery Report**, a secure full-stack engineering dashboard using **React, TypeScript**, and **FastAPI**, consolidating **Azure DevOps, GitHub**, pipeline, cloud cost, and code quality metrics into a single operational view for programme leadership.
- Implemented **Azure Key Vault**-backed secret management, **Microsoft Entra ID** token validation, backend **API proxy** with host allowlisting, and report caching with **async HTTP** for secure near real-time analytics; containerised and deployed to **AKS** via **Docker, Helm**, and **Azure Pipelines CI/CD**.
- Architected and delivered full-stack features across **Angular** micro frontends and **ASP.NET Core / .NET Core 8.0** backends using **MVC architecture** and **Entity Framework Core** with **LINQ**, integrating **RESTful Web APIs**, business logic, and **SQL Server** data layers end-to-end.
- Replaced **Duende IdentityServer** with **Microsoft Entra ID (Azure Active Directory)**, configuring app registrations, scopes, roles, and claims using **OAuth 2.0** and **OpenID Connect**; designed enterprise **RBAC** securing access across frontends, APIs, and platform features.
- Led migration from **Azure Front Door Classic** to **Azure Front Door Premium**, improving global routing, security policy enforcement, and **WAF** capabilities on **Microsoft Azure**.
- Built and optimized **CI/CD pipelines** in **Azure DevOps** using **Azure Pipelines**, automating build, test, and deployment workflows to **Azure App Services** across dev and production environments.
- Authored and optimized **SQL Server** stored procedures and data transformation logic; validated **RESTful API** contracts via **Postman, Swagger**, and **Scalar**; maintained **Cypress** end-to-end test suites to ensure backend efficiency and frontend release confidence.

EUROMONITOR INTERNATIONAL | SDE-1 | FULL STACK DEVELOPER Hybrid | Feb 2024 – Apr 2025

- Developed scalable full-stack features for the EMMA platform using **Angular, ASP.NET Core Web API, C#, Entity Framework, LINQ**, and **SQL Server**; built **.NET Web API** endpoints following the **CQRS** and **MVC** patterns for improved separation of concerns and testability.
- Improved frontend performance and responsiveness by refactoring expensive selectors with **Web Workers** and applying **responsive web design** principles with **JavaScript, jQuery**, and **CSS**; designed and optimized **SQL Server** schemas, tables, and stored procedures for data integrity and query efficiency.

- Maintained **Azure Pipelines** for automated **CI/CD** deployments; investigated and resolved defects across **Angular**, **.NET**, and **SQL Server** layers to improve platform stability.

EUROMONITOR INTERNATIONAL | SDE-1 INTERN | FULL STACK DEVELOPER INTERN Hybrid | Oct 2023 – Jan 2024

- Built e-commerce features using **Angular** and **.NET Core**, gaining hands-on experience with **RESTful Web APIs**, component-based frontend development, and Agile practices.

HACKATHONS

SPRINT RISK PREDICTOR

ML.NET, GROQ, GEMINI, AZURE DEVOPS, C#, LLM APIs

Built an ML-powered **Azure DevOps** extension predicting sprint failure risk using **ML.NET FastForest** (97.8% accuracy). Integrated **Groq Llama 3.3** and **Google Gemini 2.5 Flash** LLM APIs for role-specific AI recommendations across Devs, BAs, PMs, and Scrum Masters. Designed a 7-tab dashboard surfacing capacity imbalances, goal misalignment, and SP redistribution actions for proactive mid-sprint decisions.

PROJECTS

RESEARCHSWARM: MULTI-AGENT DEEP RESEARCH ENGINE

PYTHON, FASTAPI, LANGGRAPH,

LANGCHAIN, GROQ, REACT, TAILWIND CSS, SSE

Built a full-stack **multi-agent AI research platform** using **Python**, **FastAPI**, **LangGraph**, and **LangChain** orchestrating **Planner**, **Search**, **Reader**, **Synthesis**, and **Critic AI agents** to generate source-backed research reports end-to-end. Implemented **asyncio** concurrency and **Server-Sent Events (SSE)** for real-time streaming to a **React + Tailwind CSS** frontend; integrated **Groq LLM** and **Tavily Search API** for inference and live retrieval. Deployed on **Render** and **Vercel**.

AI SKIN SPECIALIST: MULTIMODAL HEALTHCARE AI ASSISTANT

PYTHON, FASTAPI, GEMINI VISION,

WHISPER, MULTIMODAL LLM, RAG

Built a **multimodal healthcare AI assistant** using **Python** and **FastAPI** that accepts patient **voice input**, **skin images**, and **video**; processes speech-to-text via **Whisper**, analyzes visuals with a **multimodal LLM (Gemini Vision)**, and returns structured written guidance plus synthesized **audio responses**. Designed a unified **multi-modal ingestion pipeline** handling real-world clinical input variability for scalable healthcare AI delivery.

OPEN SOURCE & COMMUNITY

AI-COOKBOOK

PYTHON, LANGCHAIN, LLAMAINDEX, OPENAI, RAG, FASTAPI

Authored a public **AI engineering reference library** with hands-on, code-first recipes covering **LLM API integration**, **RAG pipeline** construction, document extraction, and **conversational AI** patterns using **Python** — serving as a practical resource for AI practitioners.

PYDANTICAI: TYPE-SAFE LLM AGENT EXPLORATION

PYTHON, PYDANTICAI, LLM AGENTS,

STRUCTURED OUTPUTS, DEPENDENCY INJECTION

Hands-on exploration and extension of the open-source **PydanticAI** framework for building **type-safe**, production-grade **LLM agents** with structured outputs, dependency injection, tool use, and multi-step reasoning — demonstrating deep engagement with the modern AI engineering ecosystem.

CERTIFICATIONS

- **AZ-900: Microsoft Azure Fundamentals**
- **AZ-204: Microsoft Azure Developer Associate**

EDUCATION

Bachelor of Engineering in Information Science and Engineering

Bangalore, Karnataka | 2019 – 2023

RR INSTITUTE OF TECHNOLOGY, VTU | CGPA: 8.5/10

CODING PROFILES

- **LeetCode** | **TakeUForward** | **Codeforces** | **CodeChef** | **StopStalk**